

Subscription Module — Complete Documentation

Last updated: 2026-08-29 · **Scope:** Platform (SaaS) subscription — a tenant workspace subscribing to a Plan and paying the platform via Razorpay.

Status: Launch-ready. 4 launch-blocker fixes + real free tier shipped 2026-08-29. Recurring auto-billing, GST invoicing and a few hardening items are deferred — see §8.

How to use this doc in future: share `docs/subscription-module.md` (the source of this PDF) in a prompt and name the pending item from §8 you want built. Every §8 item is written so it can be picked up directly from here.

1. Two different "subscription" concepts

The codebase has **two** unrelated things both called "subscription". Do not mix them. This doc covers the **Platform Subscription** only.

	Platform Subscription (this doc)	Service Subscription
Table	<code>subscriptions</code>	<code>service_subscriptions</code>
Model	<code>App\Models\Subscription</code>	<code>App\Models\ServiceSubscription</code>
Meaning	A tenant workspace pays <i>us</i> (the SaaS) for a Plan	A tenant's <i>own customer</i> subscribes to a service the tenant sells
Money flow	Tenant → Platform (Razorpay)	Tenant's customer → Tenant (tracked, invoiced)
Gate	<code>CheckSubscription</code> middleware locks the whole workspace	<code>module:subscriptions</code> feature flag
Cron	<code>subscriptions:check-platform</code>	<code>subscriptions:auto-renew</code> , <code>subscriptions:remind-expiry</code>

2. Key files

Area	File
Model	<code>app/Models/Subscription.php</code>
Plan / Coupon models	<code>app/Models/Plan.php</code> , <code>app/Models/Coupon.php</code>
Access gate (middleware)	<code>app/Http/Middleware/CheckSubscription.php</code> — alias <code>subscription</code>
Tenant billing controller	<code>app/Http/Controllers/Web/Tenant/SubscriptionController.php</code>
Razorpay webhook	<code>app/Http/Controllers/Web/SubscriptionWebhookController.php</code>
Razorpay API wrapper	<code>app/Services/RazorpayService.php</code>
Superadmin plan CRUD	<code>app/Http/Controllers/Web/SuperAdmin/PlanController.php</code>
Superadmin manual control	<code>app/Http/Controllers/Web/SuperAdmin/TenantController.php</code> → <code>updateSubscription()</code>
Expiry / reminder cron	<code>app/Console/Commands/CheckPlatformSubscriptions.php</code>
Signup (creates trial / free sub)	<code>RegisterController.php</code> (web), <code>Api/Auth/AuthController.php</code> (API)
Routes	<code>routes/web.php</code> (search <code>subscription</code>), <code>routes/console.php</code>
Tenant views	<code>resources/views/tenant/subscription/*</code> — plans, checkout, current, success, expired
Superadmin view	<code>resources/views/superadmin/tenants/show.blade.php</code>
Notification types	<code>config/notifications.php</code>

3. Data model

subscriptions table

Column	Notes
tenant_id	FK, cascade delete
plan_id	FK to plans
coupon_id	nullable, set when a coupon is applied at checkout
original_amount, discount_amount	rupee amounts, for reporting
razorpay_order_id	created at checkout
razorpay_payment_id, razorpay_signature	set after successful payment
razorpay_subscription_id	reserved for future recurring billing — currently always null
status	enum: trial, active, cancelled, expired, past_due, pending_payment
billing_cycle	monthly yearly
trial_ends_at	when the 14-day trial ends (paid plans only)
started_at	when the current paid term started
ends_at	when the current term expires (null = free plan / no expiry)
cancelled_at	set when tenant cancels
renewal_reminder_sent_at	ADDED 2026-08-29 de-dupes the "expiring soon" email; reset to null on every activation

Tenant::subscription() = hasOne(Subscription::class)->latestOfMany() — a tenant can have many rows historically; the latest one is "current".

Status meanings

Status	Access?	Meaning
trial	Yes (until trial_ends_at)	14-day free trial on a paid plan
active	Yes (until ends_at , or forever if free)	Paid & current, or free plan
pending_payment	No	Checkout started, payment not completed
past_due	No	Payment failed
cancelled	Yes until ends_at , then No	Tenant cancelled; keeps access for the paid period
expired	No	Term ended; set by the daily cron

Model helpers (Subscription.php)

- isFree() — plan's monthly_price == 0 . Free subs never expire, never lock out.
- isActive() — status === 'active' and (free OR ends_at null OR future).
- isTrial() — status === 'trial' and trial_ends_at in the future.
- isExpired() — the gate check. Returns false for free plans. cancelled counts as expired **only after** ends_at passes.

- `daysLeft()`, `trialDaysLeft()`, `getMrrAttribute()`.

plans table

Global (no tenant scope). `features` is a JSON map used for module gating and quotas:

```
{ "leads": 500, "users": 5, "whatsapp": true, "reports": true,
  "service": true, "manufacturing": true, "subscriptions": true, ... }
```

- `-1` = unlimited.
- A `false` /absent key = feature off (superadmin can still force-on per tenant — see `Tenant::hasModuleEnabled()`).
- Seeded plans (`PlanSeeder`): **Free Trial** (₹0), **Starter** (₹999/mo), **Pro** (₹2499/mo).

Coupons

Applied at checkout via AJAX. `type` = `percentage` | `flat`, optional `max_discount` cap, `max_uses`, `expires_at`, `applicable_to` (`all` | `specific_user`). `used_count` increments **only on successful payment** (`verify()`).

4. The full flow



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Cancel — POST /subscription/cancel · ADMIN ONLY

status = cancelled, cancelled_at = now · access CONTINUES until `ends_at`, then the cron expires it



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Re-subscribe

from `/subscription/plans` or `/subscription/expired` → new `pending_payment` row → active

5. What was added on 2026-08-29

Five changes. For each: **what / why / when / how**.

5.1 Platform-subscription expiry & reminder cron DONE #2

WHAT

New command `app/Console/Commands/CheckPlatformSubscriptions.php` (`subscriptions:check-platform`), scheduled daily 07:30 in `routes/console.php`. New column `subscriptions.renewal_reminder_sent_at` (migration `2026_08_29_120000_...`). New notification types `subscription.platform_expiring` and `subscription.platform_expired` in `config/notifications.php`.

WHY

Before this, an expired workspace was still `status = active` in the DB — only `isExpired()` computed it at read-time. So (a) tenants got **zero warning** before lockout, (b) superadmin dashboards counted stale "active" subs and wrong MRR.

WHEN IT RUNS

Automatically every day at 07:30 (needs the Laravel scheduler cron — see §6). Can also be run manually any time.

HOW TO USE

```
php artisan subscriptions:check-platform          # default: warn 7 days ahead
php artisan subscriptions:check-platform --reminder-days=14
```

Output: `Platform subscriptions – expired: N, reminded: M.` Tenant admins receive in-app + email notifications; reminders link to `/subscription/plans`.

5.2 Billing routes are admin-only DONE #4

WHAT

`checkout`, `verify`, `cancel`, `apply-coupon` routes wrapped in the `tenant.admin` middleware group in `routes/web.php`. View-only routes (`plans`, `current`, `expired`, `success`) stay open to all staff.

WHY

Previously **any** staff user could purchase or **cancel the whole company's subscription**. Billing is an owner/admin action.

WHEN

Enforced on every request to those routes. A non-admin hitting them gets `403`.

HOW TO USE / EXTEND

If you add a new billing action route, put it inside the same `Route::middleware('tenant.admin')->group(...)` block in `routes/web.php` (~line 255).

5.3 Cancel no longer locks the tenant out immediately DONE #3

WHAT

`Subscription::isExpired()` now treats `cancelled` as expired **only after** `ends_at` **has passed**. Added a "cancelled — access until DATE" banner to `resources/views/tenant/subscription/current.blade.php`.

WHY

The old code returned `true` for any `cancelled` sub, so a tenant who paid for a full year and then cancelled was locked out **the same second** — contradicting the on-screen message ("you can use until ...").

WHEN

Takes effect immediately on cancel; the daily cron finishes the job when `ends_at` passes.

HOW TO USE

Nothing to do — `cancel()` already sets `cancelled_at`. The cron handles final expiry.

5.4 Real free tier DONE #5

WHAT

- `Subscription::isFree()` helper — plan `monthly_price == 0`.
- `isExpired()` returns `false` for free subs; `isActive()` returns `true` with a null `ends_at`.
- Signup (web + API) creates free-plan subs as `status = active`, `trial_ends_at = null`, `ends_at = null`.
- "You're on the free plan — no expiry" banner in `current.blade.php`; the Cancel button and "Days Remaining" row are hidden for free.

WHY

The "Free Trial" plan (₹0) used to get a 14-day `ends_at` and then lock out — and the `/expired` page only lists **paid** plans. So there was effectively **no free tier** despite one existing.

WHEN

Applies to every new free-plan signup. Existing stuck free rows are also covered because `isExpired()` now short-circuits on `isFree()`.

HOW TO USE

To make a plan behave as free, set its `monthly_price` to `0` in the superadmin Plans screen. To convert an existing tenant to free, use the superadmin manual control (below) — pick the free plan, status `active`, clear both date fields.

5.5 Superadmin manual subscription control DONE #12

WHAT

`TenantController::updateSubscription()` + route `superadmin.tenants.update-subscription` + a "Manage Subscription" form card on the tenant detail page (`superadmin/tenants/show.blade.php`).

WHY

Support had **no way** to extend a trial, move a tenant between plans, comp a plan (partnership / goodwill), or fix a stuck payment without editing the database by hand.

WHEN TO USE

- Customer paid out-of-band (bank transfer) → set plan + status `active` + `ends_at`.
- Extend a trial → bump `trial_ends_at`.
- Comp Pro for a partner → pick Pro, status `active`, set `ends_at` (or leave blank for open-ended).
- Downgrade / move to free → pick free plan, status `active`, clear dates.
- Wrongly-expired tenant → status `active` + correct `ends_at`.

HOW TO USE

Superadmin → Tenants → open a tenant → **Manage Subscription** card → set Plan / Status / Billing cycle / Expires on / Trial ends on / Note → **Save**. Every change writes an `AuditLog` row (`action = subscription_updated`) including the note.

`renewal_reminder_sent_at` is auto-reset so the "expiring soon" email can fire again for the new term.

6. Operations & setup

6.1 Environment variables (`.env`)

```
RAZORPAY_KEY_ID=rzp_live_XXXXXXX
RAZORPAY_KEY_SECRET=XXXXXXX
RAZORPAY_WEBHOOK_SECRET=XXXXXXX
```

Config: `config/services.php` → `razorpay` .

6.2 Razorpay dashboard

- Create a **Webhook** pointing at `https://<your-domain>/webhook/razorpay` with the same secret as `RAZORPAY_WEBHOOK_SECRET` .
- Subscribe to events: `payment.captured` , `payment.failed` . (`subscription.*` events are for the future recurring-billing work.)

6.3 Laravel scheduler (required for expiry & reminders)

One system cron entry runs all scheduled commands:

```
* * * * * cd /path-to-app && php artisan schedule:run >> /dev/null 2>&1
```

Verify our command is registered:

```
php artisan schedule:list | grep check-platform
# 30 7 * * * php artisan subscriptions:check-platform
```

6.4 Plans

Superadmin → Plans. Set `monthly_price = 0` for the free tier. `features` JSON drives module access and quotas. "Monthly billing enabled" toggle controls whether the public pricing page shows monthly prices.

6.5 Migration

```
php artisan migrate      # applies 2026_08_29_120000_add_renewal_reminder_sent_at_to_subscriptions_table
```

7. Testing checklist

- New signup on **free** plan → can use workspace indefinitely, no "trial ends" banner, `/subscription/current` shows "No expiry".
- New signup on **paid** plan → 14-day trial, banner counts down.
- Trial lapses (set `trial_ends_at` in the past) → next request redirects to `/subscription/expired`.
- Run `php artisan subscriptions:check-platform` → lapsed sub becomes `expired`, admin gets notification.
- Sub expiring in 5 days → command emails admin once; second run does **not** re-email.
- Checkout as a **non-admin** staff user → `403`.
- Full paid checkout in Razorpay test mode → `verify` activates, old sub cancelled, `/success` shows.
- Apply a valid coupon at checkout → amount drops, `used_count` increments only after payment.
- Cancel an active paid sub → still has access, banner shows "access until DATE", cron expires it after `ends_at`.
- Superadmin **Manage Subscription** → change plan / extend date → reflected on tenant side + `AuditLog` row created.
- Razorpay webhook with a bad signature → `400`, logged.

8. Pending work

Not done yet. Ordered by priority. Each item is self-contained — share this doc and name the item to have it built.

8.1 Recurring auto-billing HIGH LARGE #1

Problem: Only Razorpay **Orders** (one-time) are used. `razorpay_subscription_id` and the `subscription.activated/cancelled` webhook handlers exist but nothing creates a Razorpay Subscription. Every renewal is a manual re-purchase by the tenant → churn.

Build: Razorpay **Subscriptions API** with a mandate (UPI AutoPay / e-NACH / card). Store `razorpay_subscription_id`. On `subscription.charged` webhook, extend `ends_at` and record the payment. Add dunning (retry + notify) on `subscription.pending` / `halted`.

Touches: `RazorpayService` (add `createSubscription`, `cancelSubscription`), `SubscriptionWebhookController` (flesh out `subscription.*`), `SubscriptionController::checkout` (mandate flow), `Plan` (`razorpay_monthly_plan_id` / `razorpay_yearly_plan_id` already exist — wire them up).

8.2 GST tax invoice / payment receipt HIGH IF GST-REGISTERED MEDIUM #11

Problem: No downloadable invoice/receipt for platform payments. Only `razorpay_payment_id` is stored. India law requires a GST tax invoice per payment.

Build: `platform_invoices` table (number, tenant, subscription, base amount, GST %, CGST/SGST/IGST split by tenant state, total, PDF path). Generate on `verify()` + webhook activation. Download link on `/subscription/current` and superadmin tenant page. Reuse the existing invoice-PDF tooling.

8.3 Webhook / verify consistency + idempotency MEDIUM MEDIUM #7

Problem: `onPaymentCaptured` doesn't cancel old subs, doesn't set `ends_at`, doesn't bump coupon `used_count` — so browser-closed-after-payment leaves two `active` rows. No idempotency: Razorpay retries the same event.

Build: Extract the "activate this order" logic into one method called by **both** `verify()` and the webhook. Add a `processed_at` / event-id guard (or a `webhook_events` table) so repeats are no-ops.

8.4 `verify()` edge case — paid customer locked out MEDIUM SMALL #6

Problem: `verify()` cancels **all** active/trial/pending subs, then activates the row matching `razorpay_order_id`. If that row isn't found, the tenant is left with **nothing active** despite paying.

Build: Look up + validate the target row **before** cancelling anything; if not found, fetch the payment from Razorpay by `order_id` and reconstruct, or abort without cancelling. Log loudly.

8.5 Coupon hardening MEDIUM — ONLY IF RUNNING CAMPAIGNS SMALL-MEDIUM #8

- Race condition: concurrent redemptions can exceed `max_uses` (check-then-increment gap). → atomic conditional increment, or a `coupon_redemptions` table with a unique constraint.
- No tenant scoping: `Coupon.tenant_id` exists but `isValid()` ignores it — Tenant A's coupon works for Tenant B. → add the check.
- No rate limit on `apply-coupon` → coupon-code brute force. → `throttle` middleware.

8.6 `checkout` is a GET with side effects LOW SMALL #9

Problem: `GET /subscription/checkout/...` creates a DB row **and** a Razorpay Order. Prefetch / crawler / refresh creates junk orders.

Build: Split — GET renders a confirm page; `POST` creates the Order + row.

8.7 Proration on upgrade/downgrade

LOW

MEDIUM

#10

Problem: Mid-cycle plan change forfeits the remaining paid days and money; new `ends_at = now + 1 cycle`.

Build: Credit unused days of the old plan against the new plan's price at checkout; show the prorated amount.

8.8 Minor items

LOW

- `resources/views/components/layouts/navbar.blade.php:187` — `$user->tenant->subscription->status` has no null guard → page crashes if a tenant somehow has no subscription. Add `?->`.
- `resources/views/components/sidebar.blade.php` — `now()->diffInDays($trial_ends_at, false)` returns a float in Carbon 3 → shows "13.6 days left". Cast to `int`.
- Prices are cast `decimal:2` but `checkout()` casts to `(int)` → can't charge ₹499.50. Use `paise/float` end-to-end if fractional pricing is ever needed.
- `PlanController::destroy` blocks deletion when there are **active** subs but not **trial** subs → a plan with trial users can be deleted.
- `payment.failed` only sets `status = past_due` — no dunning email, no retry.
- Currency hard-coded `INR` throughout (fine while India-only).

9. Changelog

Date	Change
2026-08-29	Audit + 4 launch blockers + free tier. Added <code>subscriptions:check-platform</code> cron (expiry + 7-day reminder), <code>renewal_reminder_sent_at</code> column, <code>platform_expiring</code> / <code>platform_expired</code> notifications. Billing routes (<code>checkout</code> / <code>verify</code> / <code>cancel</code> / <code>apply-coupon</code>) restricted to <code>tenant.admin</code> . <code>isExpired()</code> fixed so <code>cancelled</code> keeps access until <code>ends_at</code> . Real free tier: <code>isFree()</code> , free signups created as permanent <code>active</code> . Superadmin <code>updateSubscription()</code> manual control with audit logging.
(earlier)	Base module: Plan CRUD, trial on signup, Razorpay Order checkout, coupon support, <code>CheckSubscription</code> gate, webhook for <code>payment.captured</code> / <code>failed</code> .